ECOM90009 Quantitative Methods for Business

Workshop 6 (Week 7)

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Introduction

Zheng Fan

- Know more about myself: zhengfan.site
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If you need any help, don't be shy

- Weekly consultation & Ed discussion board (your first priority)
- Email: `jnlye@unimelb.edu.au` for subject related issues
- Consult Stop 1 for special consideration
- Email: `fan.z@unimelb.edu.au`

Before asking any questions, make sure you have gone through the materials on Canvas and existing posts on Ed discussion board!

Introduction

- Hypothesis testing and assumptions
  - Formulating H_0 and H_A
  - One- and two-sided tests using z, t and χ^2
  - Rejection region
- Critical value and p-value approaches
- Test of single population mean and variance

Types of Error in Hypothesis Testing

	H_0 **is True**	H_0 **is False**
Fail to Reject H_0	Correct Decision	Type II Error (β)
Reject H_0	Type I Error (α)	Correct Decision (Power)

- **Type I Error:** Rejecting H_0 when it is actually true.
- **Type II Error:** Failing to reject H_0 when it is actually false.

Question 1

Indicate whether each statement below is true or false. Justify your answer.

a. A one-tail p-value is two times the size of that for a two-tail test.
b. A null hypothesis is a statement about the value of a population parameter; it is put up for testing in the face of numerical evidence.
c. A Type I error is represented by α, and is the probability of incorrectly rejecting a true null hypothesis.
d. If we do not reject the null hypothesis, we conclude that there is enough statistical evidence to infer that the null hypothesis is true.

Question 1(a)

Statement: *A one-tail p-value is two times the size of that for a two-tail test.*

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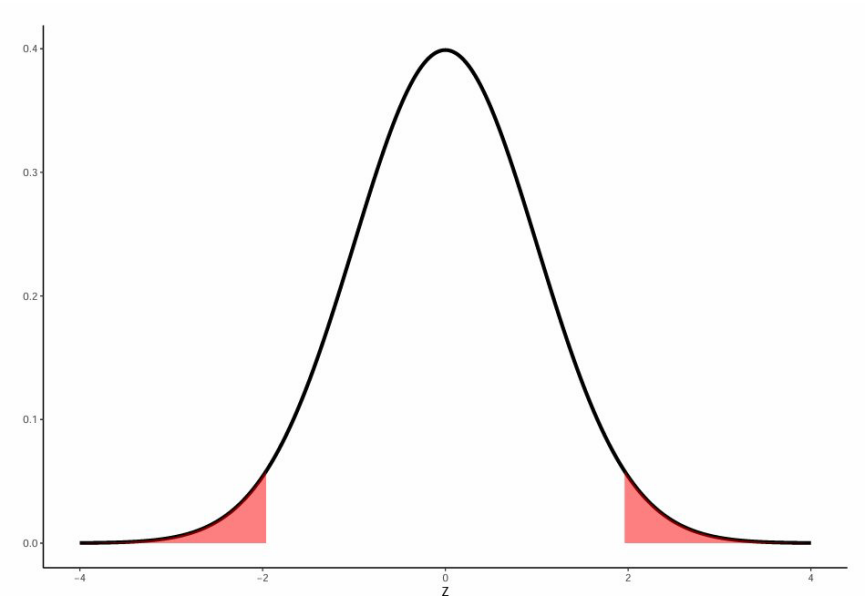
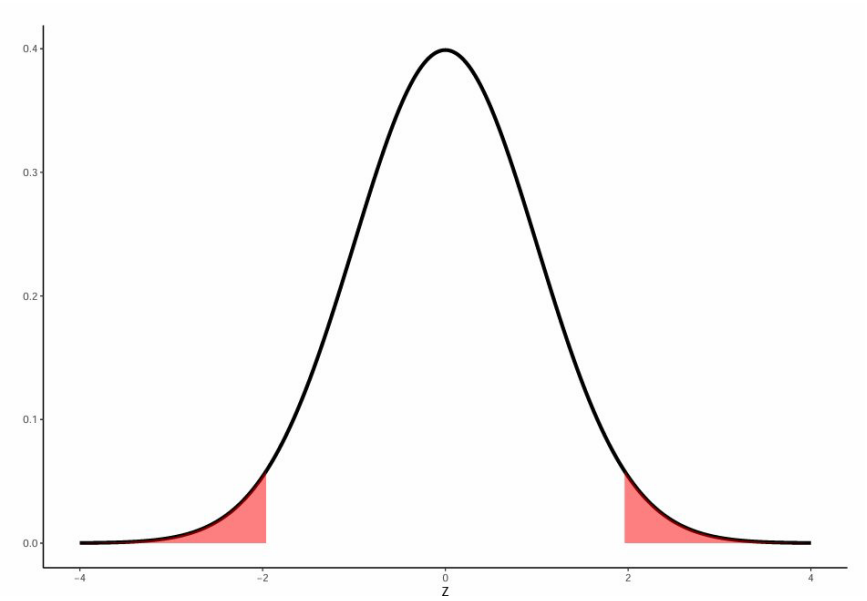
Statement: *A one-tail p-value is two times the size of that for a two-tail test.*



Question 1(a)

Statement: *A one-tail p-value is two times the size of that for a two-tail test.*



False.

- If you are using $\alpha = 0.05$, a two-tailed test divides this value in half, meaning that 0.025 is in each tail of the distribution.
- Therefore, a one-tail p-value is half the size of that for a two-tail test.

Question 1(b)

Statement: *A null hypothesis is a statement about the value of a population parameter; it is put up for testing in the face of numerical evidence.*

Question 1(b)

Statement: *A null hypothesis is a statement about the value of a population parameter; it is put up for testing in the face of numerical evidence.*

True.

- The null hypothesis is a proposition about which a decision is to be made in testing a hypothesis.

Question 1(c)

Statement: *A Type I error is represented by α, and is the probability of incorrectly rejecting a true null hypothesis.*

Question 1(c)

Statement: *A Type I error is represented by α, and is the probability of incorrectly rejecting a true null hypothesis.*

True.

- Conducting a hypothesis test always implies that there is a chance of making an incorrect decision.
- The probability of the Type I error (a true null hypothesis is rejected) is commonly called the significance level of the hypothesis test and is denoted by α.

Question 1(d)

Statement: *If we do not reject the null hypothesis, we conclude that there is enough statistical evidence to infer that the null hypothesis is true.*

Question 1(d)

Statement: *If we do not reject the null hypothesis, we conclude that there is enough statistical evidence to infer that the null hypothesis is true.*

False.

- No, we conclude that there is not enough evidence to support the alternative hypothesis.

Question 2

A critical factor in choosing a location for a new clothing store is the average clothing expenditure per household in the area. A survey of 25 randomly chosen households in a particular area yielded a mean and standard deviation of annual expenditure on clothes of \$385 and \$60 respectively.

Can you test whether, at the 5% level of significance, the population mean annual expenditure on clothing in this area is less than \$400?

If so, do so by fully completing the six steps of the test, stating explicitly:

(a) which test statistic you will use,

(b) why it is the appropriate statistic, and

(c) any additional assumptions you must make.

If you cannot conduct the test, explain why not.

Question 2 - Continued

Summary statistic of sample: $n = 25$, $\bar{x} = \$385$ and $s = \$60$

Test whether, at the 5% level of significance, the population mean annual expenditure on clothing in this area is less than \$400?

Question 2 - Continued

Summary statistic of sample: $n = 25$, $\bar{x} = \$385$ and $s = \$60$

Test whether, at the 5% level of significance, the population mean annual expenditure on clothing in this area is less than \$400?

Step 1: Is the mean annual expenditure in this area less than \$400?

$$H_0 : \mu = 400 \qquad H_A : \mu < 400$$

Question 2 - Continued

Summary statistic of sample: $n = 25$, $\bar{x} = \$385$ and $s = \$60$

Test whether, at the 5% level of significance, the population mean annual expenditure on clothing in this area is less than \$400?

Step 1: Is the mean annual expenditure in this area less than \$400?

$$H_0 : \mu = 400 \qquad H_A : \mu < 400$$

- Do we use the t-statistic or z-statistic?
- Are the expenditures normally distributed?
- What is the distribution of the test statistic?

Question 2 - Continued

Summary statistic of sample: $n = 25$, $\bar{x} = \$385$ and $s = \$60$

Question 2 - Continued

Summary statistic of sample: $n = 25$, $\bar{x} = \$385$ and $s = \$60$

Step 2: We do not know the population standard deviation, and must use the sample standard deviation instead. We are not told whether the annual clothing expenditures are normally distributed or not. If we assume that they are normally distributed, we can use the t-statistic.

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} \sim t_{n-1}$$

Question 2 - Continued

Summary statistic of sample: $n = 25$, $\bar{x} = \$385$ and $s = \$60$

Question 2 - Continued

Summary statistic of sample: $n = 25$, $\bar{x} = \$385$ and $s = \$60$

Step 3: $\alpha = 0.05$

Step 4: One-tailed (lower) test. So we reject H_0 if $t < -t_{0.05,24} = -1.711$

Question 2 - Continued

Summary statistic of sample: $n = 25$, $\bar{x} = \$385$ and $s = \$60$

Step 3: $\alpha = 0.05$

Step 4: One-tailed (lower) test. So we reject H_0 if $t < -t_{0.05,24} = -1.711$

Step 5: Calculation of test statistic

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{385 - 400}{60/\sqrt{25}} = -1.25$$

Question 2 - Continued

Summary statistic of sample: $n = 25$, $\bar{x} = \$385$ and $s = \$60$

Step 3: $\alpha = 0.05$

Step 4: One-tailed (lower) test. So we reject H_0 if $t < -t_{0.05,24} = -1.711$

Step 5: Calculation of test statistic

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{385 - 400}{60/\sqrt{25}} = -1.25$$

Step 6: Since $-1.25 > -1.711$ (not in the rejection region), we fail to reject H_0 at the 5% level of significance.

- We conclude that there is insufficient evidence that at the 5% level to support the alternative hypothesis that the population mean annual expenditure is less than \$400.

Question 2 - T-Table

Critical Values of the Student t Distribution - upper (right) tail only

Degrees of Freedom	$t_{0.10}$	$t_{0.05}$	$t_{0.025}$	$t_{0.01}$	$t_{0.005}$
1	3.078	6.314	12.706	31.821	63.657
2	1.886	2.920	4.303	6.965	9.925
3	1.638	2.353	3.182	4.541	5.841
4	1.533	2.132	2.776	3.747	4.604
5	1.476	2.015	2.571	3.365	4.032
6	1.440	1.943	2.447	3.143	3.707
7	1.415	1.895	2.365	2.998	3.499
8	1.397	1.860	2.306	2.896	3.355
9	1.383	1.833	2.262	2.821	3.250
10	1.372	1.812	2.228	2.764	3.169
11	1.363	1.796	2.201	2.718	3.106
12	1.356	1.782	2.179	2.681	3.055
13	1.350	1.771	2.160	2.650	3.012
14	1.345	1.761	2.145	2.624	2.977
15	1.341	1.753	2.131	2.602	2.947
16	1.337	1.746	2.120	2.583	2.921
17	1.333	1.740	2.110	2.567	2.898
18	1.330	1.734	2.101	2.552	2.878
19	1.328	1.729	2.093	2.539	2.861
20	1.325	1.725	2.086	2.528	2.845
21	1.323	1.721	2.080	2.518	2.831
22	1.321	1.717	2.074	2.508	2.819
23	1.319	1.714	2.069	2.500	2.807
24	1.318	1.711	2.064	2.492	2.797
25	1.316	1.708	2.060	2.485	2.787
26	1.315	1.706	2.056	2.479	2.779
27	1.314	1.703	2.052	2.473	2.771
28	1.313	1.701	2.048	2.467	2.763
29	1.311	1.699	2.045	2.462	2.756
30	1.310	1.697	2.042	2.457	2.750
35	1.306	1.690	2.030	2.438	2.724
40	1.303	1.684	2.021	2.423	2.704
45	1.301	1.679	2.014	2.412	2.690
50	1.299	1.676	2.009	2.403	2.678

Question 3

Summary statistic of sample: $n = 25$, $\bar{x} = \$385$ and $s = \$60$

Using the information provided in Question 2 above, can you test whether, at the 5% level of significance, the population variance for annual expenditure in this area is different from $\$57^2$?

If so, do so by fully completing the six steps of the test, stating explicitly:

(a) which test statistic you will use,

(b) why it is the appropriate statistic, and

(c) any additional assumptions you must make.

Question 3 - Continued

Summary statistic of sample: $n = 25$, $\bar{x} = \$385$ and $s = \$60$

Test whether, at the 5% level of significance, the population variance for annual expenditure in this area is different from $\$57^2$?

Question 3 - Continued

Summary statistic of sample: $n = 25$, $\bar{x} = \$385$ and $s = \$60$

Test whether, at the 5% level of significance, the population variance for annual expenditure in this area is different from $\$57^2$?

Step 1: Is the population variance for annual expenditure in this area different from $\$57^2$?

$$H_0 : \sigma^2 = 57^2 \qquad H_A : \sigma^2 \neq 57^2$$

Question 3 - Continued

Summary statistic of sample: $n = 25$, $\bar{x} = \$385$ and $s = \$60$

Test whether, at the 5% level of significance, the population variance for annual expenditure in this area is different from $\$57^2$?

Step 1: Is the population variance for annual expenditure in this area different from $\$57^2$?

$$H_0 : \sigma^2 = 57^2 \qquad H_A : \sigma^2 \neq 57^2$$

- What test statistic do we use?
- Are the expenditures normally distributed?
- What is the distribution of the test statistic?

Question 3 - Continued

Summary statistic of sample: $n = 25$, $\bar{x} = \$385$ and $s = \$60$

Question 3 - Continued

Summary statistic of sample: $n = 25$, $\bar{x} = \$385$ and $s = \$60$

Step 2: We are not told whether the annual clothing expenditures are normally distributed or not. If we assume that they are normally distributed, we can use the χ^2-statistic.

$$\chi^2 = \frac{(n-1)s^2}{\sigma^2} \sim \chi^2_{n-1}$$

Question 3 - Continued

Summary statistic of sample: $n = 25$, $\bar{x} = \$385$ and $s = \$60$

Question 3 - Continued

Summary statistic of sample: $n = 25$, $\bar{x} = \$385$ and $s = \$60$

Step 3: $\alpha = 0.05$

Step 4: Two-tailed test. So we reject H_0 if $\chi^2 < \chi^2_{0.975,24} = 12.4$ or if $\chi^2 > \chi^2_{0.025,24} = 39.4$

Question 3 - Continued

Summary statistic of sample: $n = 25$, $\bar{x} = \$385$ and $s = \$60$

Step 3: $\alpha = 0.05$

Step 4: Two-tailed test. So we reject H_0 if $\chi^2 < \chi^2_{0.975,24} = 12.4$ or if $\chi^2 > \chi^2_{0.025,24} = 39.4$

Step 5: Calculation of test statistic

$$\chi^2 = \frac{(n-1)s^2}{\sigma^2} = \frac{24 \times 60^2}{57^2} = 26.59$$

Question 3 - Continued

Summary statistic of sample: $n = 25$, $\bar{x} = \$385$ and $s = \$60$

Step 3: $\alpha = 0.05$

Step 4: Two-tailed test. So we reject H_0 if $\chi^2 < \chi^2_{0.975,24} = 12.4$ or if $\chi^2 > \chi^2_{0.025,24} = 39.4$

Step 5: Calculation of test statistic

$$\chi^2 = \frac{(n-1)s^2}{\sigma^2} = \frac{24 \times 60^2}{57^2} = 26.59$$

Step 6: Since $12.4 < 26.59 < 39.4$ (not in the rejection region), we fail to reject H_0 at the 5% level of significance.

- We conclude that there is insufficient evidence that at the 5% level to support the alternative hypothesis that the population variance for annual expenditure is different from 57^2.

Question 3 - Chi-Squared Table

Critical values of the Chi-squared distribution χ^2 - upper (right) tail

d.f.	$\chi^2_{0.995}$	$\chi^2_{0.99}$	$\chi^2_{0.975}$	$\chi^2_{0.95}$	$\chi^2_{0.90}$	$\chi^2_{0.10}$	$\chi^2_{0.05}$	$\chi^2_{0.025}$	$\chi^2_{0.01}$	$\chi^2_{0.005}$
1	0.000039	0.000157	0.000982	0.00393	0.0158	2.71	3.84	5.02	6.63	7.88
2	0.01	0.02	0.05	0.10	0.21	4.61	5.99	7.38	9.21	10.6
3	0.07	0.11	0.22	0.35	0.58	6.25	7.81	9.35	11.34	12.8
4	0.21	0.30	0.48	0.71	1.06	7.78	9.49	11.1	13.28	14.9
5	0.41	0.55	0.83	1.15	1.61	9.24	11.1	12.8	15.1	16.7
6	0.68	0.87	1.24	1.64	2.20	10.6	12.6	14.4	16.8	18.5
7	0.99	1.24	1.69	2.17	2.83	12.0	14.1	16.0	18.5	20.3
8	1.34	1.65	2.18	2.73	3.49	13.4	15.5	17.5	20.1	22.0
9	1.73	2.09	2.70	3.33	4.17	14.7	16.9	19.0	21.7	23.6
10	2.16	2.56	3.25	3.94	4.87	16.0	18.3	20.5	23.2	25.2
11	2.60	3.05	3.82	4.57	5.58	17.3	19.7	21.9	24.7	26.8
12	3.07	3.57	4.40	5.23	6.30	18.5	21.0	23.3	26.2	28.3
13	3.57	4.11	5.01	5.89	7.04	19.8	22.4	24.7	27.7	29.8
14	4.07	4.66	5.63	6.57	7.79	21.1	23.7	26.1	29.1	31.3
15	4.60	5.23	6.26	7.26	8.55	22.3	25.0	27.5	30.6	32.8
16	5.14	5.81	6.91	7.96	9.31	23.5	26.3	28.8	32.0	34.3
17	5.70	6.41	7.56	8.67	10.1	24.8	27.6	30.2	33.4	35.7
18	6.26	7.01	8.23	9.39	10.9	26.0	28.9	31.5	34.8	37.2
19	6.84	7.63	8.91	10.1	11.7	27.2	30.1	32.9	36.2	38.6
20	7.43	8.26	9.59	10.9	12.4	28.4	31.4	34.2	37.6	40.0
21	8.03	8.90	10.3	11.6	13.2	29.6	32.7	35.5	38.9	41.4
22	8.64	9.54	11.0	12.3	14.0	30.8	33.9	36.8	40.3	42.8
23	9.26	10.2	11.7	13.1	14.8	32.0	35.2	38.1	41.6	44.2
24	9.89	10.9	12.4	13.8	15.7	33.2	36.4	39.4	43.0	45.6
25	10.5	11.5	13.1	14.6	16.5	34.4	37.7	40.6	44.3	46.9
26	11.2	12.2	13.8	15.4	17.3	35.6	38.9	41.9	45.6	48.3
27	11.8	12.9	14.6	16.2	18.1	36.7	40.1	43.2	47.0	49.6
28	12.5	13.6	15.3	16.9	18.9	37.9	41.3	44.5	48.3	51.0
29	13.1	14.3	16.0	17.7	19.8	39.1	42.6	45.7	49.6	52.3
30	13.8	15.0	16.8	18.5	20.6	40.3	43.8	47.0	50.9	53.7
35	17.2	18.5	20.6	22.5	24.8	46.1	49.8	53.2	57.3	60.3
40	20.7	22.2	24.4	26.5	29.1	51.8	55.8	59.3	63.7	66.8

Question 4

A survey of 100 people shows that 42 enjoy bike riding.

(a) Can you test the hypothesis that less than 50% of the population enjoy bike riding at the 5% level of significance?

If so, do so by fully completing the six steps of the test, stating explicitly:

(i) which test statistic you will use,

(ii) why it is the appropriate statistic, and

(iii) any additional assumptions you must make.

If you cannot conduct the test, explain why not.

(b) If you tested this hypothesis using the p-value approach, would we expect to get the same result? Explain why or why not.

Question 4a - critical value approach

Test the hypothesis that less than 50% of the population enjoy bike riding at the 5% level of significance?

Question 4a - critical value approach

Test the hypothesis that less than 50% of the population enjoy bike riding at the 5% level of significance?

- Let X and $\hat{p}$ be the number and sample proportion of people who enjoy bike riding respectively.

$$\hat{p} = \frac{X}{n} = \frac{42}{100} = 0.42$$

Question 4a - critical value approach

Test the hypothesis that less than 50% of the population enjoy bike riding at the 5% level of significance?

- Let X and $\hat{p}$ be the number and sample proportion of people who enjoy bike riding respectively.

$$\hat{p} = \frac{X}{n} = \frac{42}{100} = 0.42$$

<u>Step 1:</u> Do less than 50% of the population enjoy bike riding?

$$H_0 : p = 0.5 \qquad H_A : p < 0.5$$

Question 4a - critical value approach

Test the hypothesis that less than 50% of the population enjoy bike riding at the 5% level of significance?

- Let X and $\hat{p}$ be the number and sample proportion of people who enjoy bike riding respectively.

$$\hat{p} = \frac{X}{n} = \frac{42}{100} = 0.42$$

<u>Step 1:</u> Do less than 50% of the population enjoy bike riding?

$$H_0 : p = 0.5 \qquad H_A : p < 0.5$$

- What is the test statistic?
- What is the distribution of the test statistic?

Question 4a - Continued

A survey of 100 people shows that 42 enjoy bike riding.

Question 4a - Continued

A survey of 100 people shows that 42 enjoy bike riding.

<u>Step 2:</u> Given $np = 100 \times 0.5 = 50 > 5$ and $nq = 100 \times 0.5 = 50 > 5$, the test statistic

$$z = \frac{\hat{p} - p}{\sqrt{pq/n}}$$

has an approximate standard normal distribution.

Question 4a - Continued

A survey of 100 people shows that 42 enjoy bike riding.

Question 4a - Continued

A survey of 100 people shows that 42 enjoy bike riding.

Step 3: $\alpha = 0.05$

Step 4: One-tailed (lower) test. So we reject H_0 if $z < -z_{0.05} = -1.645$

Question 4a - Continued

A survey of 100 people shows that 42 enjoy bike riding.

Step 3: $\alpha = 0.05$

Step 4: One-tailed (lower) test. So we reject H_0 if $z < -z_{0.05} = -1.645$

Step 5: Calculation of test statistic

$$z = \frac{\hat{p} - p}{\sqrt{pq/n}} = \frac{0.42 - 0.5}{\sqrt{0.5 \times 0.5/100}} = -1.6$$

Question 4a - Continued

A survey of 100 people shows that 42 enjoy bike riding.

Step 3: $\alpha = 0.05$

Step 4: One-tailed (lower) test. So we reject H_0 if $z < -z_{0.05} = -1.645$

Step 5: Calculation of test statistic

$$z = \frac{\hat{p} - p}{\sqrt{pq/n}} = \frac{0.42 - 0.5}{\sqrt{0.5 \times 0.5/100}} = -1.6$$

Step 6: Since $-1.6 > -1.645$ (not in the rejection region), we fail to reject H_0 at the 5% level of significance.

- We conclude that there is insufficient evidence that at the 5% level to support the alternative hypothesis that less than 50% of the population enjoy bike riding.

Question 4b - using p-value approach

A survey of 100 people shows that 42 enjoy bike riding.

Test the hypothesis that less than 50% of the population enjoy bike riding at the 5% level of significance?

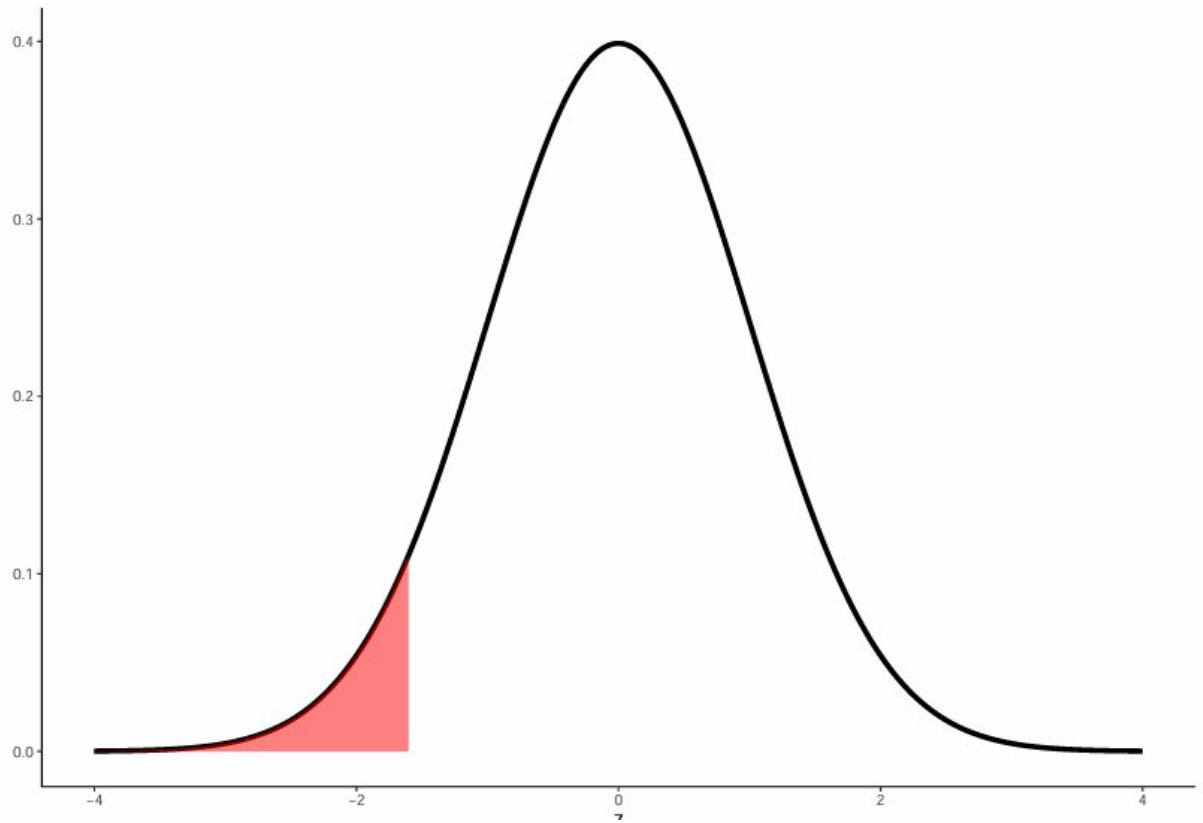
$$H_0 : p = 0.5 \qquad\qquad H_A : p < 0.5$$

Calculated test statistic from part (a):

$$z = \frac{\hat{p} - p}{\sqrt{pq/n}} = \frac{0.42 - 0.5}{\sqrt{0.5 \times 0.5/100}} = -1.6$$

p-value $= P(Z < -1.6)$

Question 4b - Visual Representation



Would the conclusion change if we change the significance level to 10% instead of 5%?

Thanks for your attention!

Let me know if there is any questions!